SGA 7-I

Exposé IX. Néron Models and Monodromy

Monodromy Pairings and Component Groups

9. Definition of the monodromy pairing

9.1. The setting

In this section the trait S is henselian and the abelian variety A_K has semi-stable reduction over S. Equivalently, the same is true for the dual abelian variety A'_K. We shall use the twisted constant $\mathbb{Z}$-modules M and M' attached to the maximal tori of the special fibres of the connected Neron models of A_K and A'_K. After a suitable finite etale extension of S these become constant free groups, non-canonically isomorphic to $\mathbb{Z}^r$, where r is the common rank of the toric parts.

For each prime ℓ, put

$$M_\ell = M \otimes \mathbb{Z}_\ell, \qquad M'_\ell = M' \otimes \mathbb{Z}_\ell. \tag{9.1.1}$$

The geometric meaning of these sheaves is given by the formulae of the preceding sections, where they occur as the Tate modules of the toric parts. Our aim is to define a canonical bilinear form, the *monodromy pairing*,

$$u_\ell : M_\ell \otimes M'_\ell \longrightarrow \mathbb{Z}_\ell. \tag{9.1.2}$$

It will be defined by means of the Barsotti-Tate pro-ℓ-groups $T_\ell(A^0)$, $T_\ell(A'^0)$, and even by means of the generic fibre $T_\ell(A_K)$ of the first of these groups. When $\ell = p$, this last description uses Tate's theorem for homomorphisms of Barsotti-Tate groups; for the present purposes it is available in the equalities of characteristic to which it is applied below.

When $\ell \neq p$, the same construction may be expressed in purely Galois-theoretic terms. If

$$U = T_\ell(A_K(\bar{K})) \tag{9.1.3}$$

with its action of the inertia group I, then u_ℓ is the datum which measures the unipotent monodromy of I on U.

9.2. The case $\ell \neq p$

Assume first that $\ell \neq p$. We use the notation introduced earlier for the exact sequence of I-modules attached to the semi-stable reduction of A_K. Up to torsion, the relation which defines the toric submodule says that it is generated by elements of the form

$$gx - x \qquad (x \in U,\ g \in I). \tag{9.2.1}$$

Thus for each $g \in I$ one has

$$gx = x + u(g)(x), \tag{9.2.2}$$

where $u(g)$ is a homomorphism from U to the toric part, vanishing on that toric part. Equivalently,

$$u(g) \in \mathrm{Hom}(U/V, V), \tag{9.2.3}$$

where V denotes the submodule on which the transvections are parallel. In the semi-stable case the operation of every g is a transvection parallel to V, and composition of such transvections corresponds to addition of the associated homomorphisms. Hence the action of inertia is described by a homomorphism

$$I \longrightarrow \mathrm{Hom}(U/V, V). \tag{9.2.4}$$

By the orthogonality theorem applied both to A_K and to A'_K, one identifies U/V with the Tate twist of the toric Tate module of A'_K. Consequently the homomorphism (9.2.4) may be interpreted as a canonical homomorphism

$$M_\ell \otimes M'_\ell \longrightarrow \mathbb{Z}_\ell, \tag{9.2.5}$$

or as an element

$$u_\ell \in \mathrm{Hom}(M_\ell \otimes M'_\ell, \mathbb{Z}_\ell). \tag{9.2.6}$$

This is the element announced in (9.1.2) in the case $\ell \neq p$. It is invariant under transport of structure, hence is defined over the original residue field and not merely after passing to a separably closed residue field.

To define (9.1.2) without excluding $\ell = p$, we next recall a purely homological construction.

9.3. Mixed extensions

Let $\mathcal{C}$ be an abelian category and let

$$P, \quad R, \quad Q \tag{9.3.1}$$

be three objects of $\mathcal{C}$. We are interested in objects X of $\mathcal{C}$ endowed with a three-step filtration

$$0 = X^3 \subset X^2 \subset X^1 \subset X$$

with specified identifications of the successive quotients

$$X/X^1 \simeq Q, \qquad X^1/X^2 \simeq R, \qquad X^2 \simeq P.$$

Writing

$$F = X^1, \qquad E = X/X^2,$$

one obtains two ordinary extensions

$$0 \to P \to F \to R \to 0, \qquad 0 \to R \to E \to Q \to 0. \tag{9.3.2}$$

Conversely, suppose that the two extensions F and E in (9.3.2) are fixed. A *mixed extension* of E by F, or in Grothendieck's terminology an *extension panachee*, is such a filtered object X, together with isomorphisms

$$X^1 \simeq F, \qquad X/X^2 \simeq E, \tag{9.3.3}$$

compatible with the filtrations and with the common quotient R.

For fixed P, Q, R, E, F, the mixed extensions of E by F form a groupoid, denoted

$$\mathrm{EXTPAN}(E, F).$$

Every arrow is invertible, and the automorphism group of any object X is canonically

$$\mathrm{Hom}(Q, P). \tag{9.3.4}$$

Indeed an endomorphism of the underlying object of X which is the identity on the graded pieces differs from the identity by the composite

$$X \longrightarrow Q \longrightarrow P \longrightarrow X.$$

The groupoid $\mathrm{EXTPAN}(E, F)$ is naturally a torsor under the category-group $\mathrm{EXT}(Q, P)$ of extensions of Q by P. The action is the mixed analogue of the Baer sum. If G is an extension of Q by P and X is a mixed extension of E by F, then

$$(G, X) \longmapsto G \wedge X \tag{9.3.5}$$

is obtained by applying the Baer composition to the corresponding extensions of Q by F. For each fixed X, the functor

$$G \longmapsto G \wedge X$$

from $\mathrm{EXT}(Q, P)$ to $\mathrm{EXTPAN}(E, F)$ is an equivalence, and one has the associativity relation

$$G \wedge (G' \wedge X) = (G \wedge G') \wedge X. \tag{9.3.7}$$

It follows that

a) $\mathrm{EXTPAN}(E, F)$ is a groupoid, and the automorphism group of each object is canonically $\mathrm{Ext}^0(Q, P) = \mathrm{Hom}(Q, P)$;

b) the set $\mathrm{Extpan}(E, F)$ of isomorphism classes is either empty or is a torsor under $\mathrm{Ext}^1(Q, P)$;

c) there is a canonical obstruction class

$$e(E, F) \in \mathrm{Ext}^2(Q, P), \tag{9.3.9}$$

whose vanishing is necessary and sufficient for the existence of a mixed extension of E by F.

The class is described as follows. A mixed extension of E by F is equivalently an extension of E by P whose image in $\mathrm{Ext}^1(R, P)$ is the class of F. In the exact sequence obtained from the second sequence in (9.3.2),

$$\mathrm{Ext}^1(E, P) \longrightarrow \mathrm{Ext}^1(R, P) \xrightarrow{\partial} \mathrm{Ext}^2(Q, P),$$

one has

$$e(E, F) = \partial([F]). \tag{9.3.10}$$

The construction is functorial for exact functors. If

$$r : \mathcal{C} \longrightarrow \mathcal{C}' \tag{9.3.11}$$

is exact, and if primes denote images under r, then there is a commutative diagram of groupoids, up to canonical isomorphism,

$$\begin{array}{ccc} \mathrm{EXT}(Q, P) \times \mathrm{EXTPAN}(E, F) & \longrightarrow & \mathrm{EXTPAN}(E, F) \\ \downarrow & & \downarrow \\ \mathrm{EXT}(Q', P') \times \mathrm{EXTPAN}(E', F') & \longrightarrow & \mathrm{EXTPAN}(E', F'). \end{array} \tag{9.3.12}$$

Assume $\mathrm{EXTPAN}(E, F) \neq \varnothing$ and choose $X \in \mathrm{EXTPAN}(E, F)$. For a mixed extension Y' of E' by F', one obtains a class

$$e(Y') \in \mathrm{Ext}^1(Q', P')/\Im\, \mathrm{Ext}^1(Q, P). \tag{9.3.13}$$

It is independent of the choice of X, and vanishes precisely when Y' is isomorphic to the image by r of a mixed extension of E by F.

9.4. A finite-level specialization criterion

Let S be a henselian trait with generic point η, let $n > 0$, and put $\Lambda = \mathbb{Z}/n\mathbb{Z}$. Let $\mathcal{C}$ be the category of Λ-modules on S, for the fpqc topology, and let $\mathcal{C}_\eta$ be the corresponding category on η. Let P, R, Q be objects of $\mathcal{C}$, with E an extension of Q by R and F an extension of R by P. Assume that Q is locally free of finite type over Λ_S, and that P is represented by a finite multiplicative-type group over S, say

$$P = D(Q^*), \tag{9.4.0}$$

where D denotes Cartier duality. Suppose a mixed extension X_η of E_η by F_η is given. We shall attach to it a canonical pairing

$$e(X_\eta) : Q_\eta \otimes Q_\eta^* \longrightarrow \Lambda_\eta. \tag{9.4.1}$$

If S is strictly local, then Q and Q^* are constant, and giving (9.4.1) is the same as giving an element of

$$Q^\vee \otimes (Q^*)^\vee. \tag{9.4.2}$$

Apply the construction of 9.3 to the restriction functor $\mathcal{C} \to \mathcal{C}_\eta$. The obstruction group $\operatorname{Ext}^2(Q, P)$ vanishes, so mixed extensions over S exist. The required identifications rest on the following computations.

Lemma 9.4.3. Under the preceding hypotheses, with S strictly local, one has canonical isomorphisms

$$\begin{aligned}\operatorname{Ext}^i(Q, P) &\simeq H^i(S, Q^* \otimes P),\\ \operatorname{Ext}^i(Q_\eta, P_\eta) &\simeq H^i(\eta, Q^* \otimes P),\end{aligned}$$

and

$$Q^* \otimes P \simeq D(Q \otimes Q^*).$$

For every finite constant group H on S,

$$H^i(S, D(H)) = H^i(\eta, D(H)) = 0 \quad (i \geq 2),$$

and there is a canonical isomorphism

$$H^1(\eta, D(H))/\Im H^1(S, D(H)) \simeq H^\vee.$$

The proof is obtained from Cartier duality and the Kummer sequence. In the case $H = \mathbb{Z}/m\mathbb{Z}$, the cohomology sequence associated with

$$0 \longrightarrow \mu_m \longrightarrow \mathbb{G}_m \xrightarrow{m} \mathbb{G}_m \longrightarrow 0$$

reduces the assertion to the vanishing of the relevant cohomology of $\mathbb{G}_m$ over a strictly henselian trait and over its generic point. This also describes explicitly the boundary homomorphism giving the last isomorphism.

Combining these computations with 9.3 gives a canonical isomorphism

$$\operatorname{Ext}^1(Q_\eta, P_\eta)/\Im \operatorname{Ext}^1(Q, P) \simeq Q \otimes Q^*. \tag{9.4.5}$$

The class attached in 9.3 to the generic mixed extension X_η therefore becomes the pairing (9.4.1). Its vanishing is exactly the obstruction to extending X_η to a mixed extension over S. The construction over a merely henselian trait is obtained by descent from the strict henselization.

Proposition 9.4.7. With the same data, without assuming that the trait is henselian, the restriction functor

$$\mathrm{EXTPAN}(E,F) \longrightarrow \mathrm{EXTPAN}(E_\eta, F_\eta)$$

is fully faithful. Its essential image consists of those generic mixed extensions for which the pairings (9.4.1), at all closed points after henselization, vanish.

The proof is local for the etale topology on S. Full faithfulness follows because homomorphisms extend uniquely from the generic point to S, using normality and finiteness of the Cartier-dual group. Essential surjectivity is reduced to the strictly local case, where it is precisely the obstruction computation above.

9.5. Mixed extensions of Barsotti-Tate pro-groups

Fix a prime ℓ. Let

$$P, \quad R, \quad Q \tag{9.5.1}$$

be pro-ℓ-Barsotti-Tate groups on S, and suppose given extensions

$$0 \to P \to F \to R \to 0, \qquad 0 \to R \to E \to Q \to 0. \tag{9.5.2}$$

For every $n \geq 0$ this gives finite-level extensions

$$0 \to P(n) \to F(n) \to R(n) \to 0, \qquad 0 \to R(n) \to E(n) \to Q(n) \to 0. \tag{9.5.3}$$

Assume Q is of etale type and P of multiplicative type. If over η a pro-ℓ-Barsotti-Tate group X_η is given as a mixed extension of E_η by F_η, then each finite level $X_\eta(n)$ gives a pairing of the form

$$Q(n) \otimes Q^*(n) \longrightarrow \mathbb{Z}/\ell^{n+1}\mathbb{Z}. \tag{9.5.4}$$

These pairings are compatible in n, hence define a homomorphism of twisted constant ℓ-adic sheaves

$$e(X_\eta) : Q \otimes Q^* \longrightarrow \mathbb{Z}_\ell. \tag{9.5.5}$$

Proposition 9.5.6. The restriction functor from mixed Barsotti-Tate extensions over S to those over η is fully faithful. Its essential image consists of the generic mixed extensions for which the pairing (9.5.5) vanishes. If $\operatorname{char} K = 0$, so that Tate's theorem on homomorphisms of Barsotti-Tate groups applies, then a generic mixed extension belongs to the essential image as soon as its underlying Barsotti-Tate group extends over S.

The first assertion follows level by level from Proposition 9.4.7. For the last assertion, extend the generic Barsotti-Tate group X_η to X over S. Tate's theorem extends uniquely the homomorphism $X_\eta \to Q_\eta$. Its finite-level maps are epimorphisms, so their kernels form a Barsotti-Tate group F, and the generic isomorphism with the prescribed F_η extends. The quotient by P is similarly identified with E, giving a mixed extension over S.

When $\ell \neq p$, all these Barsotti-Tate groups are equivalently twisted constant ℓ-adic sheaves, or representations of $\mathrm{Gal}(\bar{K}/K)$ on finite free $\mathbb{Z}_\ell$-modules. The homomorphism (9.5.5) is then exactly the homomorphism measuring the inertia action, as described in 9.2.

9.6. Definition in the semi-stable case

We now apply the preceding construction to the semi-stable abelian variety A_K. Let

$$P = T_\ell(T), \qquad Q = M', \qquad R = T_\ell(A^0), \tag{9.6.1}$$

and let F be the resulting extension of R by P. The dual extension attached to A'_K, after Cartier duality and the orthogonality theorem, gives an extension E of Q by R. As generic mixed extension one takes

$$X_\eta = T_\ell(A_K), \tag{9.6.2}$$

with its canonical three-step filtration induced by the toric and fixed parts. We are therefore in the situation of 9.5, and the pairing (9.5.5) takes precisely the form (9.1.2). For $\ell \neq p$ it agrees with the inertia construction of 9.2. Under Tate's theorem, it may be described entirely in terms of the Barsotti-Tate pro-ℓ-group $T_\ell(A_K)$ over the generic point.

9.7. Descent from a finite semi-stable extension

If A_K is not assumed semi-stable, choose a finite Galois extension K'/K such that $A_{K'}$ has semi-stable reduction over the normalization S' of S. Descending the monodromy pairing from K', and normalizing by the ramification index

$$e(S'/S) = \mathrm{length}(V'/\mathfrak{m}V'),$$

one obtains a pairing

$$u_\ell : M_K \otimes M'_K \longrightarrow \mathbb{Q}_{\ell,K} \tag{9.7.1}$$

which is independent of the chosen semi-stable extension. When A_K already has semi-stable reduction, it is the restriction to the generic point of (9.1.2). The integrality theorem below shows that this pairing is in fact independent of ℓ and has values in $\mathbb{Q}_K$. In the elliptic curve case the pairing can be interpreted as an ordinary rational number; Neron's results identify it with the valuation of the absolute invariant j.

10. Properties of the monodromy pairing. Integrality and positivity

This section records the formal properties of the monodromy pairings and states the deeper integrality and positivity theorem. The verifications of the formal properties are straightforward from the definitions and are left mostly to the reader.

10.1. Transport of structure

The definition of (9.1.2) is compatible with transport of structure in the triple (S, A_K, A'_K). Thus every isomorphism of such data carries the corresponding modules M, M' and the pairing u_ℓ to the corresponding objects for the transported data.

10.2. Symmetry

Apply the construction to the dual abelian variety A'_K. Its dual is canonically identified with A_K. The resulting pairing

$$u_\ell^{A'} : M'_\ell \otimes M_\ell \longrightarrow \mathbb{Z}_\ell \tag{10.2.1}$$

is obtained from u_ℓ^A by the transposition isomorphism

$$M'_\ell \otimes M_\ell \xrightarrow{\sim} M_\ell \otimes M'_\ell.$$

In other words,

$$u_\ell^{A'}(x', x) = u_\ell^A(x, x'). \tag{10.2.2}$$

10.3. Change of trait

Let $S' \to S$ be a dominant morphism of traits, and let $A_{K'}$ be obtained from A_K by base change. The monodromy pairing changes by multiplication with the ramification index:

$$u_\ell^{S'} = e(S'/S)\, u_\ell^S. \tag{10.3.4}$$

This is compatible with the description by inertia when $\ell \neq p$, and in the general case follows by the finite-level mixed extension construction.

Theorem 10.4. Let S be a henselian trait and let A_K have semi-stable reduction over S. Let M and M' be the character lattices introduced in 9.1. Then:

a) There exists a unique pairing

$$u : M \otimes M' \longrightarrow \mathbb{Z} \tag{10.4.1}$$

whose ℓ-adic extension is the pairing u_ℓ of (9.1.2), for every prime ℓ. This pairing is separating.

b) If $\xi : A_K \to A'_K$ is a polarization, and if

$$(x, y) = u(x, \xi y) \tag{10.4.2}$$

denotes the induced bilinear form on M, then this form is symmetric, positive, and non-degenerate.

Equivalently, after choosing a geometric point of S, the induced form on the fibre M_0 is a symmetric positive non-degenerate integral form. The pairing (10.4.1) is again called the monodromy pairing.

10.5. Reduction of the proof

The uniqueness in Theorem 10.4 is immediate, and existence may be checked after any dominant change of traits. It is therefore enough to prove the theorem after replacing S by a complete strictly local trait. If A_K is isogenous to a direct factor of an abelian variety for which the theorem is known, then the theorem is known for A_K; this follows from functoriality of the lattices and of the monodromy pairing, together with Poincare reducibility.

After a finite extension of K, every abelian variety is a quotient of the Jacobian of a proper smooth geometrically connected curve. By a theorem of Artin, after another finite extension the curve has a regular proper flat model whose geometric special fibre has only ordinary quadratic singularities, i.e. normal crossings of two branches. Thus Theorem 10.4 is reduced to the case of such a Jacobian. The integral formula and positivity will then be verified geometrically in the Picard-Lefschetz calculation of Section 12.

It remains to check that positivity for one polarization implies positivity for every polarization. If ξ_0 and ξ are two polarizations, their associated homomorphisms satisfy, in $\mathrm{End}(A_K) \otimes \mathbb{Q}$,

$$\xi = \xi_0 a = a^* \xi_0, \qquad a = b^* b$$

for some $b \in \mathrm{End}(A_K) \otimes \mathbb{Q}$. In terms of the biextensions attached to the polarizations, this gives

$$\mathcal{B}(\xi) = \mathcal{B}(\xi_0) \circ (b, b). \tag{10.5.5.1}$$

Consequently

$$u_\xi(x, y) = u_{\xi_0}(bx, by). \tag{10.5.5.2}$$

Symmetry and positivity are therefore inherited from ξ_0, and non-degeneracy is equivalent to the separating property of (10.4.1), which is independent of the polarization.

11. Connected components of the Neron model: duality and asymptotic behavior

11.0. Component groups

Let

$$\Phi = A/A^0 \tag{11.0.1}$$

be the finite etale group of connected components of the Neron model. For each prime ℓ, let $\Phi(\ell)$ be its ℓ-primary component. The knowledge of Φ is equivalent to the knowledge of all the $\Phi(\ell)$. Since formation of the Neron model is compatible with strict henselization, the study of Φ may be reduced to the strictly local case; the Galois action on the component group is then recovered by transport of structure.

11.1. Strictly local description

Assume S strictly local. Then

$$\Phi(k) \simeq A(k)/A^0(k) \simeq A(S)/A^0(S) \simeq A(K)/A(K)^0. \tag{11.1.1}$$

Here the first equality uses smoothness over the residue field, the second is Hensel's lemma for the smooth group scheme A over S, and the third is the equality $A(S) = A(K)$ for the Neron model. We put

$$A(K)^0 = A^0(S). \tag{11.1.2}$$

For n prime to $p = \mathrm{char}\, k$, multiplication by n is etale and surjective on A^0, hence $A(K)^0$ is n-divisible. Therefore

$$\Phi(k)/n\Phi(k) \simeq A(K)/nA(K). \tag{11.1.3}$$

If $\ell \neq p$, let

$$U = T_\ell(A_K(\bar{K})). \tag{11.1.5}$$

Since the fixed part of A_K is described by the inertia invariants of U, (11.1.3) yields, for n a power of ℓ,

$$\Phi(k)/n\Phi(k) \simeq U_n^I/(U^I)_n. \tag{11.1.6}$$

Passing to the direct limit over n gives:

Proposition 11.2. Assume S strictly local and $\ell \neq p$. There is a canonical isomorphism

$$\Phi(\ell) \simeq (U \otimes D_\ell)^I/(U^I \otimes D_\ell), \qquad D_\ell = \mathbb{Q}_\ell/\mathbb{Z}_\ell. \tag{11.2.1}$$

Thus the ℓ-primary component group is determined by the Tate module U and the inertia action.

11.3. The duality pairing for $\ell \neq p$

Let $U' = T_\ell(A'_K(\bar{K}))$, and let

$$U \otimes U' \longrightarrow T_\ell(\mathbb{G}_m) \tag{11.3.2}$$

be the perfect Weil pairing, compatible with the inertia action. The structure of inertia gives an exact sequence

$$1 \longrightarrow R \longrightarrow I \longrightarrow T \longrightarrow 1 \tag{11.3.3}$$

with R pro-prime-to-ℓ. Put

$$\Phi = (U \otimes D_\ell)^I/(U^I \otimes D_\ell), \quad \Phi' = (U' \otimes D_\ell)^I/(U'^I \otimes D_\ell). \tag{11.3.4}$$

We define a perfect pairing

$$\Phi \otimes \Phi' \longrightarrow \mathbb{Q}_\ell/\mathbb{Z}_\ell. \tag{11.3.5}$$

Consider the exact sequence

$$0 \to U \to U \otimes \mathbb{Q}_\ell \to U \otimes D_\ell \to 0 \tag{11.3.6}$$

and the corresponding cohomology exact sequence for I. The group Φ is canonically identified with the kernel of the connecting map into $H^1(I,U)$, hence with the torsion subgroup of $H^1(I,U)$. The same holds for U'. The standard duality for the cohomology of I, using the perfect pairing (11.3.2), gives perfect pairings

$$H^0(I, U \otimes D_\ell) \times H^1(I, U')_{\mathrm{tor}} \longrightarrow D_\ell, \tag{11.3.12}$$

and after passage to the limit,

$$H^0(I, U \otimes D_\ell) \times H^1(I, U' \otimes D_\ell) \longrightarrow D_\ell. \tag{11.3.13}$$

The maps occurring in the two exact sequences are transposes of one another under these pairings. Therefore the cokernel on one side and the kernel on the other are canonically dual, which gives the pairing (11.3.5).

11.4–11.6. The semi-stable formula for all primes

The preceding description is only for $\ell \neq p$. Suppose now that A_K has semi-stable reduction. For every prime ℓ we shall define perfect pairings

$$\Phi(\ell) \otimes \Phi'(\ell) \longrightarrow \mathbb{Q}_\ell/\mathbb{Z}_\ell. \tag{11.4.1}$$

This is done by expressing $\Phi(\ell)$ in terms of the Barsotti-Tate group $T_\ell(A_K)$ and the monodromy pairing.

Theorem 11.5. Assume S henselian and A_K semi-stable. Let ℓ be any prime, and let

$$u_\ell : M_\ell \otimes M'_\ell \longrightarrow \mathbb{Z}_\ell$$

be the monodromy pairing. Let

$$\overline{u}_\ell : M_\ell \longrightarrow \mathrm{Hom}(M'_\ell, \mathbb{Z}_\ell)$$

be the corresponding homomorphism, and put $\Phi_u(\ell) = \operatorname{coker} \overline{u}_\ell$. Then there is a canonical isomorphism

$$\Phi(\ell) \simeq \Phi_u(\ell). \tag{11.5.1}$$

In particular $\overline{u}_\ell$ is an isogeny, including for $\ell = p$. After Theorem 10.4 one may equivalently state the result by means of the integral monodromy pairing $u : M \otimes M' \to \mathbb{Z}$: if Φ_u is the cokernel of $M \to \mathrm{Hom}(M', \mathbb{Z})$, then

$$\Phi \simeq \Phi_u. \tag{11.5.3}$$

To construct the isomorphism, we reduce to the strictly local case. Let T and T' be the maximal tori, and identify

$$M = D(T), \qquad M' = D(T'). \tag{11.6.1}$$

It is enough to define, for each power n of ℓ, compatible isomorphisms

$$\Phi(n) \simeq \mathrm{Ker}(M_n \xrightarrow{\overline{u}_n} M_n'^{\vee}). \tag{11.6.4}$$

Let A_n be the kernel of multiplication by n on the Neron model. It is flat and quasi-finite over S. Its fixed and toric parts give a canonical filtration

$$A_n \supset A_n^f \supset A_n^t \supset 0. \tag{11.6.5}$$

The quotient

$$Y = A_n^f / A_n^t \tag{11.6.6}$$

is flat, separated, and quasi-finite, with generic fibre identified with $(A_K^0)_n^f / (A_K^0)_n^t$. The formulas from the preceding section give a canonical isomorphism on the generic fibre

$$M_n \xrightarrow{\sim} Y_\eta. \tag{11.6.7}$$

Lemma 11.6.8. The group scheme Y is etale over S, and (11.6.7) extends uniquely to a morphism

$$M_n \longrightarrow Y \tag{11.6.8.1}$$

which is an open immersion.

The etaleness follows from the special fibre; normality and separatedness extend the morphism, and an etale monomorphism is an open immersion.

Let

$$\Lambda_n = Y \cap A_n^0. \tag{11.6.9}$$

It is a finite open subgroup of Y, and the composite

$$\Lambda_n \hookrightarrow Y \hookleftarrow M_n \tag{11.6.11}$$

identifies Λ_n with a subgroup of M_n. It is enough to prove:

Lemma 11.6.13. The image of Λ_n in M_n is exactly the kernel of $\overline{u}_n$.

The kernel is the left annihilator of the finite-level monodromy pairing

$$M_n \otimes M_n' \longrightarrow \mathbb{Z}/n\mathbb{Z},$$

which was defined in terms of the two finite extensions and of the generic mixed extension. By Proposition 9.4.7, a subgroup belongs to this annihilator precisely when the corresponding pulled-back mixed extension extends over S. The subgroup Λ_n has such an extension by construction, so $\Lambda_n \subset \mathrm{Ker}\,\overline{u}_n$. Conversely, if a subgroup $K_n \subset \mathrm{Ker}\,\overline{u}_n$ is given, the associated mixed extension extends over S. The induced morphism into A_n factors through A_n^0, hence through Λ_n; this gives $K_n \subset \Lambda_n$. The lemma follows, and with it Theorem 11.5.

Remarks 11.7. The preceding construction shows that the finite mixed extensions (11.6.5), and hence the subgroup $A_n^0 \subset A_n$, can be reconstructed from the structural data of 9.6 and from the generic mixed extension. The annihilator condition determines the open finite subgroup, and gluing along the common open part recovers the extension. Thus the system of finite mixed extensions is encoded by the Barsotti-Tate mixed extension and the monodromy pairing.

11.8–11.10. Asymptotic behavior under finite extensions

Let $\bar{K}$ be the separable closure of K, written as the filtered union of finite extensions K'. For each K', let $S[K']$ be the normalization of S in K', and let

$$\Phi[K'] \tag{11.8.1}$$

be the component group of the Neron model of $A_{K'}$. After a sufficiently large finite extension the reduction is semi-stable, and the transition maps on component groups are injective. Put

$$\Psi = \varinjlim_{K'} \Phi[K']. \tag{11.8.2}$$

For each prime ℓ, Theorem 11.5 gives, for large K', an identification

$$\Phi[K'](\ell) \simeq \operatorname{Ker}(M \otimes D_\ell \xrightarrow{u(K')} M' \otimes D_\ell). \tag{11.8.3}$$

The transition maps are compatible with the inclusions into $M \otimes D_\ell$; hence one obtains a canonical monomorphism

$$\Psi \hookrightarrow M \otimes \mathbb{Q}/\mathbb{Z}. \tag{11.8.5}$$

Theorem 11.9. The homomorphism (11.8.5) is an isomorphism.

It remains only to prove surjectivity. The image is described by the kernels in (11.8.3), and the behavior of the monodromy pairing under ramified extension is given by (10.3.4). Thus it is enough to use the elementary fact that, over a strictly local trait, every integer can be made to divide the ramification degree of a suitable finite separable extension. This makes every element of $M \otimes \mathbb{Q}/\mathbb{Z}$ appear at some finite stage.

Corollary 11.10. Under the hypotheses in which the semi-stable reduction and monodromy constructions apply, the group of connected components of the model after passage through all finite extensions has the form

$$\varinjlim_{K'} \Phi[K'] \simeq M \otimes \mathbb{Q}/\mathbb{Z}. \tag{11.10.1}$$

For $\ell \neq p$ this is already immediate from the inertia description of Proposition 11.2. The delicate part is the p-primary component, where Theorem 11.5 is needed. The existence of the isomorphism in Corollary 11.10, with the right-hand side understood as the group associated with the toric character lattice, was conjectured by J.-P. Serre.